Operational Research

5CM522



5CM522 Assessment Brief – Coursework

Part B

Dr Sam O’Neill

Sensitivity: Internal

Operational Research (5CM522)

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Module Leader

- Dr Sam O'Neill
- Contact via the module page

Key dates and details

Assessment Type:	Individual
Assessment weighting:	25%
Learning Outcomes:	1, 2
Submission Method:	Electronic Submission via Course Resources and GitHub Classroom
Submission Date:	8th May 2026 – 12:00 (Midday/Noon)
Provisional Feedback Release Date:	29th May 2026

Description of the assessment

Students will be required to model and answer a scenario-based problem.

Module Learning Outcomes

1. Demonstrate an understanding of Operational Research methods and methodology.
2. Build models to enable the analysis and solution of case study problems.

Assessment Regulations

The University's regulations, policies and procedures for students define the framework within which teaching, and assessment are conducted. Please make sure you are familiar with these regulations, policies, and procedures.

Please make sure you are familiar with the regulations on plagiarism and academic misconduct.

This assignment is individual and you are not to work with or share answers with other students.

Assessment Rubric

Please see the assessment rubric section.

Assessment AI-assistance

- You may use generative AI tools (e.g. ChatGPT, Copilot, Claude) to support your work.
- You may not use agent-based AI tools.
- You must remain fully responsible for all modelling decisions, code, and interpretations.
- You must be able to explain and justify your work in your notebook and video presentation.
- Where there is a significant mismatch between technical sophistication and demonstrated understanding, a viva voce examination may be required.
- You need to acknowledge how AI tools were used. Please either keep a document of prompts or export/link to a chat history. These should be submitted as evidence of your usage.
- You must not submit AI-generated content that you do not understand or cannot explain.
- Any misuse of generative AI tools could be considered ethical academic misconduct (as per Academic Regulations, Academic Misconduct, Section J2). If in doubt, please consult your module leader.

Assessment Learning outcomes

By the end, students should be able to:

- Build and validate a predictive model for area-level demand.
- Formulate and solve a facility-location problem with opening costs.
- Integrate ML outputs into an optimisation model and run scenario/sensitivity analysis.
- Communicate results succinctly to a non-technical stakeholder.

Part B: Demand-Based Facility Location Brief

1. Overview of the Assessment

This coursework requires you to design, implement, and interpret a data-driven optimisation model for a realistic business problem. You will integrate machine learning forecasting with an appropriate optimisation or decision-making approach (e.g. ILP, heuristics, simulation, or hybrid methods).

A decision model may be mathematical, algorithmic, heuristic, or simulation-based, provided it is clearly defined and justified.

The primary aim is to assess your ability to:

- model real-world operational problems,
- integrate predictive analytics with optimisation,
- interpret results in a business context,
- communicate recommendations clearly to non-technical stakeholders.

2. Scenario: Yodafone Retail Reopening Optimisation

Yodafone is one of the UK's largest telecommunications and electronics retailers, with over 300 physical stores nationwide. It sells mobile devices, accessories, and broadband packages, and provides in-store technical support and bill payment facilities.

Like many high-street businesses, Yodafone was forced to close a large proportion of its stores during a recent nationwide disruption. As restrictions eased, senior management needed to decide **which stores to reopen first** and how to reallocate resources to meet customer demand efficiently.

Yodafone's challenge is complex:

- **Customer behaviour has shifted** — some customers are visiting stores less frequently, while others are travelling further for in-person service.
- **Online sales have grown**, but certain services (repairs, hands-on device demos, complex account changes) still require a store visit.
- **Operating costs remain high** — staffing, utilities, and lease agreements make it unprofitable to keep low-traffic stores open.
- **Competitor presence** has changed, with some rival stores closing permanently and others expanding.

The company's strategic goal is to **maximise total profit** during the phased reopening by selecting an optimal set of stores to operate, while ensuring that key regions remain served.

Profit should be defined consistently as revenue minus all associated costs.

3. Your role

You are a data analyst in Yodafone's Strategic Operations Team. Your task is to design a two-stage decision process:

Stage 1 — Demand Forecasting

You will estimate post-reopening demand for each geographic zone using historical sales and socio-economic data.

Stage 2 — Network Optimisation

You will formulate and solve an optimisation model to determine:

- which stores to open,
- how demand should be allocated to open stores,
- the resulting profit, costs, and trade-offs.

Your model must account for:

- fixed opening costs,
- service and distance costs,
- store capacities,
- demand allocation constraints.

4. Available data

Yodafone has provided:

- **zones.csv** — demographic, economic, and competitor data for 500 geographic zones, with historical sales figures.
- **stores.csv** — location, capacity, and fixed opening costs for 100 candidate stores.
- **zone_store_dist.csv** — distances between each zone and each store.
- **margins.json** — economic parameters, including average margin per unit sold and service cost per km.

You will work with a personalised version of the dataset.

NOTE: You may not be required to use all of the columns in the dataset, some may not be appropriate for the scenario, but were included as part of the dataset, this is common in the real-world.

5. Assessment Artefacts

The assessment consists of two components:

The Jupyter Notebook (80%)

The Jupyter Notebook (`notebook.ipynb`) is the **primary assessed artefact**. It should contain the modelling workflow, results, and explanation.

Marking will primarily evaluate:

- modelling correctness,
- analytical reasoning,
- integration of forecasting and optimisation,
- interpretation of results,
- clarity of explanation.

Your notebook must:

- run end-to-end without manual intervention,
- include both code and explanatory markdown,
- contain labelled plots and tables,
- clearly link technical results to business implications.

Slides and Video (20%)

These assess the ability to communicate the analysis and recommendation to a non-technical stakeholder.

A business-focused summary addressing:

- the decision,
- the recommendation,
- the value created,
- trade-offs and risks.

6. Deliverables

You must submit the following three components:

1. Completed Jupyter Notebook (notebook.ipynb)

- Executes from top to bottom (Kernel → Restart & Run All).
- Contains structured sections and commentary.
- Includes final conclusions and recommendations.

Submitted via GitHub Classroom.

2. Executive Slide Deck (5–6 slides, PDF)

- Please see the "Exec slide deck guidance" at the end of this brief.

Submitted via Blackboard.

3. Recorded Video Presentation (10 ± 1 minutes)

- Narrated walkthrough of slides and key decisions.
- Must cover: problem, recommendation, value, scenarios, risks, and next steps.

Over- or under-time submissions may be penalised.

Submitted via Blackboard

4. Acknowledgement of how AI tools were used

- Acknowledge how AI tools were used. Please either keep a document of prompts or export/link to a chat history. These should be submitted as evidence of your usage.

Submitted via Blackboard

7. Notebook Structure and Milestones

Your notebook must be organised into the following milestones.
Each milestone must include code, results, and explanatory commentary.

Milestone 1 — Demand Forecasting

- Feature selection and justification.
- Model choice and rationale.
- Validation approach (e.g., hold-out, cross-validation).
- Metrics (R^2, MAE/RMSE) and interpretation.
- Limitations and uncertainty.

Milestone 2 — Economic Baselines

- All-stores-open baseline.
- Single-store or simplified baseline.
- Evidence of cannibalisation and inefficiency.

Milestone 3 — Optimisation Model

- Decision variables, parameters, and constraints.
- Objective function (profit maximisation).
- Solver choice and correctness checks.
- Explanation of modelling assumptions.

Milestone 4 — Solution Analysis

- Which stores open and why.
- Profit versus baseline.
- Capacity utilisation and geographic patterns.
- Business sense check of results.

Milestone 5 — Scenario and Sensitivity Analysis

- At least two scenarios (e.g., demand changes, cost changes, capacity constraints).
- Comparison of outcomes.
- Discussion of robustness and trade-offs.

Modelling Reflection (Short Section)

Include a short reflection (approximately **150–250 words**) discussing:

- **One modelling assumption** you made and why it was reasonable in the context of the problem.
- **One limitation** of your modelling approach or data that could affect the results.
- **One improvement or extension** you would implement if more time, data, or resources were available.

Your reflection should refer to **specific aspects of your model or results**, rather than general comments about modelling or data science.

Summary and Recommendations

Provide a short executive-style summary of your analysis (**approximately 200–300 words**).

Your summary should include:

- **Your recommended store network** (e.g. how many stores open).
- **The estimated profit or value created** compared with a baseline.
- **Key trade-offs or insights** from the optimisation results.
- **One or two risks or uncertainties** decision-makers should be aware of.

This section should focus on **decision insight and interpretation**, not technical detail.

Assessment Rubric

Marks will be awarded for the quality of reasoning and justification, not the complexity of tools used.

The notebook milestones correspond broadly to the rubric criteria.

Demand Modelling and Forecasting (20%)

Mark Range	Descriptor
70–100% (First)	Model choice is justified; features are sensible; validation is rigorous; metrics are interpreted critically; limitations are discussed.
60–69% (2:1)	Appropriate model and validation; metrics reported with reasonable interpretation.
50–59% (2:2)	Basic model with limited justification and interpretation.
40–49% (Third)	Weak or poorly justified modelling approach.
0–39% (Fail)	Incorrect or meaningless forecasting.

Decision Modelling and Optimisation (20%)

Mark Range	Descriptor
70–100% (First)	Clear formulation of objectives, variables, and constraints; method is appropriate and justified; results are validated and economically coherent.
60–69% (2:1)	Mostly clear formulation; reasonable method; minor gaps in logic or justification.
50–59% (2:2)	Basic decision logic; method works but is simplistic or partially justified.
40–49% (Third)	Ad hoc or poorly defined decision model.
0–39% (Fail)	No meaningful decision model.

Integration, Scenarios and Insight (20%)

Mark Range	Descriptor
70–100% (First)	Forecast feeds decision model meaningfully; multiple scenarios analysed; trade-offs clearly reasoned; strong business insight.
60–69% (2:1)	Forecast integrated; at least two scenarios; reasonable interpretation.
50–59% (2:2)	Basic integration; limited scenario depth.
40–49% (Third)	Minimal integration or superficial insight.
0–39% (Fail)	No meaningful integration or scenario analysis.

Understanding and Justification (15%)

Mark Range	Descriptor
70–100% (First)	Clear, coherent explanation of modelling choices and results in notebook.
60–69% (2:1)	Mostly clear explanations with minor gaps.
50–59% (2:2)	Descriptive rather than analytical explanation.
40–49% (Third)	Weak or inconsistent understanding.
0–39% (Fail)	Inability to explain own work.

Technical Quality and Reproducibility (5%)

Mark Range	Descriptor
70–100% (First)	Notebook executes using Restart & Run All without manual intervention; clear structure; labelled outputs; clean code.
60–69% (2:1)	Mostly reproducible with minor issues.
50–59% (2:2)	Runs with small fixes; cluttered structure.
40–49% (Third)	Hard to run or poorly organised.
0–39% (Fail)	Non-reproducible submission.

Executive Communication (Slides and Video) (20%)

Mark Range	Descriptor
70–100% (First)	Clear and concise executive presentation. Recommendation is well supported by evidence. Business value, trade-offs, and risks are communicated effectively.
60–69% (2:1)	Good presentation of recommendation and results. Communication is mostly clear with minor gaps in explaining value or trade-offs.
50–59% (2:2)	Basic presentation of results. Recommendation or value is not communicated clearly or slides contain unnecessary technical detail.
40–49% (Third)	Weak or unclear presentation. Recommendation is poorly supported or difficult to follow.
0–39% (Fail)	Slides/video missing, incoherent, or no clear recommendation presented.

Exec slide deck Guidance (5–6 slides)

Your slide deck should answer these five questions in order:

1. **Slide 1: Problem & Opportunity**

Briefly explain the decision facing Yodafone and why it matters.
Focus on business context rather than technical detail.

2. **Slide 2: The Recommendation**

Present your recommended store network.
This may include a table, map, or summary of which stores should open or close.

3. **Slide 3: The Value**

Explain the benefits of your recommendation compared to a baseline (e.g. all stores open).
Focus on profit, coverage, and operational efficiency.

4. **Slide 4: Trade-offs & Scenarios**

Demonstrate how your recommendation changes under alternative assumptions (e.g. demand or cost changes).
Highlight key trade-offs rather than technical mechanics.

5. **Slide 5: Risks & Next Steps**

Identify key uncertainties and propose practical next steps.

Optional Slide 6: Appendix / Q&A

Optional technical detail or supporting evidence.

The examples above are illustrative rather than prescriptive. Your slides should summarise the key reasoning and insights developed in your notebook.